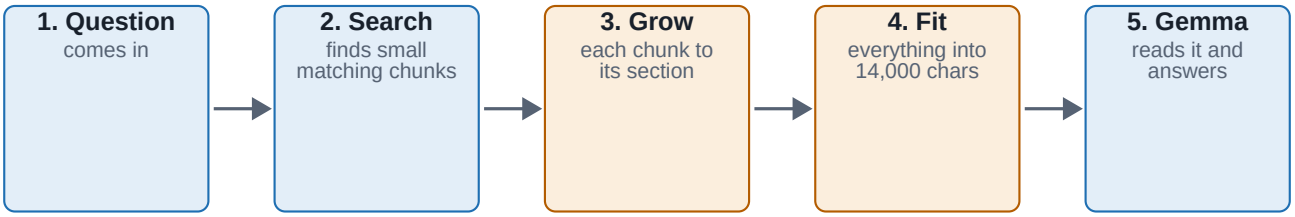
How Gemma's Reading Pack Is Built

Team 1 RAG agent: what text the model reads before it answers, and how much.
Plain-language guide with real examples from the indexed papers.

The big picture

Gemma never reads whole papers. For each question the agent collects a small **reading pack** of up to **14,000 characters** (about 3,500 tokens, roughly 4 to 5 pages) and Gemma answers only from that pack.



Steps 3 and 4 decide how much context Gemma gets. The rest of this guide explains them.

Step 3: grow each chunk into its section

The search works on small chunks of a few hundred characters, because small pieces are easier to match. A small chunk on its own often misses the sentence around the answer, so every chunk that passes the search is grown back into the section it came from.

The rule, in one line

If the whole section fits in 2,400 characters, Gemma gets the whole section. If it is longer, Gemma gets a 2,400-character window that starts 350 characters before the matched chunk.

Short section: LLaMA "2.3 Optimizer"

Long section: LoRA "5.1 Baselines"

The window starts a little before the match so the sentence that introduces it is kept. Most sections are short (the typical one is about 580 characters), but the long ones hold two-thirds of all the text.

The four cases, with real examples

Case	When	What Gemma gets	Real example
A. Short section	The section fits (2,400 chars or less)	The whole section	"What optimizer does LLaMA use?" found *2.3 Optimizer*: 462 chars, kept whole
B. Long section	The section is longer than 2,400 chars	A 2,400-char window around the match	"Which weight matrices does LoRA adapt?" found *5.1 Baselines*: 3,642 chars, kept 2,400
C. Match not found	The chunk's text cannot be located in its section	Only the small chunk	A safety fallback. In today's index it happens 0 times out of 6,455 chunks
D. No section	The text is not from a paper (study guide notes)	Only the small chunk	Only for questions that name no paper

Real mix: "How many tokens was GPT-3 trained on?" got 5 short sections (case A, about 3,300 chars in total). "Which weight matrices does LoRA adapt?" got 1 short section and 4 full windows (about 10,500 chars). Questions about methods land in long sections and use more of the pack.

Step 4: fit everything into 14,000 characters

Two limits stop the pack from growing too big.

Limit	Rule	Why
Per search	At most 5 windows, one per section. Stop when less than 400 characters of room is left.	Keeps one search from repeating the same section
Per question	14,000 characters shared equally between the parts of the question.	Every part of a multi-part question gets a fair share
No cutting	A window that does not fit is skipped, never cut short.	Cutting could remove the very sentence that matched

Example: a question with two parts
"Which base models do **ReAct** and **Toolformer** build on?" becomes two searches, one per paper. Each gets **7,000 characters** (14,000 / 2). The parts take turns adding windows, so ReAct cannot fill the whole pack before Toolformer gets a turn. With 5 parts, each part gets 2,800 characters: room for only one full window.

Two other ways the agent adds context

These change what is **searched for**, not what Gemma reads.

Helper	When it runs	Example
Question context	A question is split into several parts about the same paper, and a part no longer names the paper	Part "What learning rate was used?" is searched together with the full question "What batch size and learning rate does LLaMA use?"

Helper	When it runs	Example
Abbreviation meaning	The question has a short capital-letter name that a paper defines	"PPO" is searched together with "Proximal Policy Optimization", found in a paper that writes "Proximal Policy Optimization (PPO)"

How to give Gemma more context

Four settings control the size of the pack. Change them together.

Setting	File	Today	What it controls
CONTEXT_CHARACTERS	app/retrieval.py	14,000	Total size of the reading pack
Window size	app/retrieval.py	2,400	Largest piece taken from one section
EVIDENCE_COUNT	app/retrieval.py	5	Most windows per search
num_ctx	app/llm.py	8,192 tokens	How much text Gemma can hold at once

Two things to check before raising it

1. Gemma's own limit. The pack (about 3,500 tokens), the instructions (about 1,000) and the longest answer (3,200) must all fit in num_ctx. A bigger pack needs a bigger num_ctx, otherwise the start of the prompt is silently dropped.

2. Speed. Reading the pack is the slowest step on a CPU. A pack 40% bigger makes each answer roughly 40% slower, and the final test stops each question at 120 seconds.

Words used in this guide

Word	Meaning
Chunk	A small piece of a paper (a few hundred characters) used for searching
Section	A whole part of a paper under one heading, such as "2.3 Optimizer"
Window	The piece of a long section that Gemma reads: 2,400 characters around the match
Reading pack	Everything Gemma reads for one question: at most 14,000 characters
Token	The unit Gemma counts text in: about 4 characters of English